Matrix - DiscountCombinationsAndMarginProtection - Engine

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Document ID:     Matrix - DiscountCombinationsAndMarginProtection - Engine
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Matrix (Algorithmic Evaluation & Margin Governance Engine)
Purpose:         DiscountCombinationsAndMarginProtection
Tier / Level:    Engine (Mathematical Permutations & Stacking Policy)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Lead TechOps Specialists, E-Commerce Growth Directors, Financial Analysts
Ecosystem Scope: Shopify Admin (Discounts Engine), Shopify Checkout API, Carrier Rate Cards, Contribution Margin
Modeling
Business Unit:   Synapse Botanicals (Commercial Strategy & Margin Protection)
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1. Executive Summary & Promotional Unit Economics

Promotional discounting is one of the most effective revenue drivers in e-commerce, but unconstrained or misconfigured discounting is the fastest driver of margin collapse. In modern multi-node DTC operations, shipping rates, warehouse pick/pack fees, payment processing tolls, and product Cost of Goods Sold (COGS) are fixed physical cash outflows. If marketing promotions stack indiscriminately at checkout, order contribution margins can instantly turn negative.

This document serves as the **authoritative mathematical decision engine** for Synapse Botanicals. It establishes:

1. **The 3x3 Discount Class Permutation Matrix:** A combinatorial ruleset regulating every interaction between Product, Order, and Shipping discount classes.
2. **The Contribution Margin Protection Floor:** Mathematical models establishing the strict **35% minimum contribution margin threshold** across all catalog SKUs.
3. **Cart Evaluation Hierarchy & Conflict Resolution:** The deterministic sequence by which Shopify's calculation engine applies competing promotions.
4. **Diagnostic Triage Response Trees:** Fast-path troubleshooting workflows for resolving "Code Rejected at Checkout" customer escalations.

> **NOTE**
> **System Architecture Dependency:**
> This decision engine operationalizes the diagnostic protocols established in **SOP - TechnicalDiagnosisAndAuditLogging - Parent (Domain 5.1)** – **Section 5.0** and relies directly on the carrier zone postage baselines defined in **Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)** – **Section 2.0**.

2. The 3x3 Discount Class Permutation & Combination Matrix

Shopify separates discounts into three discrete classes: **Product Discounts** (line-item level), **Order Discounts** (cart subtotal level), and **Shipping Discounts** (delivery rate level). Each discount rule contains explicit "*Combines with*" configuration checkboxes.

The matrix below governs all permitted, conditional, and strictly prohibited permutations across the Synapse Botanicals storefront:

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Permutation Taxonomy:
• [PERMITTED]  : Approved for standard storefront operations.
• [CONDITIONAL]: Requires specific order threshold ($50.00+) or margin verification.
• [PLATFORM-X] : Technically blocked by Shopify core architecture.
• [POLICY-X]   : Prohibited by Synapse Botanicals financial governance.
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Primary Applied Class	Secondary Candidate Class	Combination Setting in Admin	Shopify Platform Behavior	Synapse Botanicals Governance Policy	Operational Rationale
Product Discount *(e.g., 20% Off Sitewide)*	**Product Discount** *(e.g., Buy 1 Get 1 50% Off)*	Checkbox: `Product discounts`	Stacks multiplicatively on the same line item or across lines.	**[POLICY-X] PROHIBITED** *(Must leave box unchecked)*	Prevents compounding line-item discounts (e.g., 20% sitewide + 25% VIP SKU discount = 40% margin destruction).
Product Discount *(e.g., 20% Off Sitewide)*	**Order Discount** *(e.g., WELCOME15)*	Checkbox: `Order discounts`	Applies Order discount to the *remaining* subtotal after Product discount.	**[CONDITIONAL]** Permitted ONLY if order subtotal $> \$75.00$ post-discount.	Stacking a sitewide sale with an introductory code on small baskets drives contribution margin below the 35% floor.
Product Discount *(e.g., 20% Off Sitewide)*	**Shipping Discount** *(e.g., FREESHIP)*	Checkbox: `Shipping discounts`	Applies Product discount to items; zeroes out carrier shipping rate.	**[CONDITIONAL]** Permitted ONLY if cart contains ≥ 1 Stick Pack or Tincture.	Free shipping is approved if the remaining paid basket can absorb regional shipping costs ($4.85–$6.85).
Order Discount *(e.g., WELCOME15)*	**Order Discount** *(e.g., INFLUENCER10)*	Checkbox: `Order discounts`	**Not supported by Shopify.**	**[PLATFORM-X] BLOCKED** Shopify allows exactly **one** order-level discount code per checkout.	Shopify core engine enforces single order-level code. Secondary code entry automatically overwrites or rejects.
Order Discount *(e.g., WELCOME15)*	**Product Discount** *(e.g., Sale Collection)*	Checkbox: `Product discounts`	Deducts product discount first; deducts order discount from residual.	**[CONDITIONAL]** Permitted during routine non-sale periods only.	During active sitewide promotions (Q4 / Black Friday), all Order discounts must have this box unchecked.
Order Discount *(e.g., WELCOME15)*	**Shipping Discount** *(e.g., FREESHIP)*	Checkbox: `Shipping discounts`	Deducts percentage/dollar from subtotal; waives shipping fee.	**[PERMITTED]** Standard promotional acquisition stack.	Order discount + Free Shipping is the standard acquisition bundle for carts over $50.00.
Shipping Discount *(e.g., FREESHIP)*	**Shipping Discount** *(e.g., Carrier Promo)*	Checkbox: `Shipping discounts`	**Not supported by Shopify.**	**[PLATFORM-X] BLOCKED** Shopify permits only **one** shipping discount per checkout.	A delivery rate can only be discounted once. Multiple shipping discounts cannot stack.
Shipping Discount *(e.g., FREESHIP)*	**Product Discount** *(e.g., 10% Off Tinctures)*	Checkbox: `Product discounts`	Applies line discount; waives shipping fee.	**[PERMITTED]** Standard category promotion stack.	Category promotions on high-margin botanicals easily absorb ground shipping costs.

Primary Applied Class	Secondary Candidate Class	Combination Setting in Admin	Shopify Platform Behavior	Synapse Botanicals Governance Policy	Operational Rationale
Shipping Discount *(e.g., FREESHIP)*	**Order Discount** *(e.g., $10 Off $50)*	Checkbox: `Order discounts`	Deducts order discount; waives shipping fee.	**[PERMITTED]** Threshold-qualified stack.	Safe if minimum order qualification is $\geq \$50.00$.

3. Contribution Margin Protection Floor & Mathematical Modeling

To prevent promotions from eroding unit economics, Synapse Botanicals mandates a **Hard 35% Contribution Margin Floor ($CM \geq 35\%$)** across all orders.

3.1 The Order Contribution Margin Formula

$$\text{Net Revenue} = \text{Gross Merchandise Value (GMV)} - \text{Product Discounts} - \text{Order Discounts}$$

$$\text{Total Fulfillment Cost} = \text{Product COGS} + \text{3PL Handling Fee} + \text{Packaging Corrugate} + \text{Carrier Postage}$$

$$\text{Gateway Fee} = (\text{Net Cash Collected} \times 2.9\%) + \$0.30$$

$$\text{Contribution Margin (\$)} = \text{Net Cash Collected} - \text{Total Fulfillment Cost} - \text{Gateway Fee}$$

$$\text{Contribution Margin (\%)} = \left(\frac{\text{Contribution Margin (\$)}}{\text{Net Revenue}} \right) \times 100$$

3.2 SKU Unit Economics & Maximum Allowable Discount Thresholds

The table below models the exact unit economics across all five production products, calculating the **Maximum Allowable Blended Discount** before contribution margin breaches the 35% floor (assuming Transcon Zone 7–8 shipping from `LOC-NV-02` or `LOC-PA-01`):

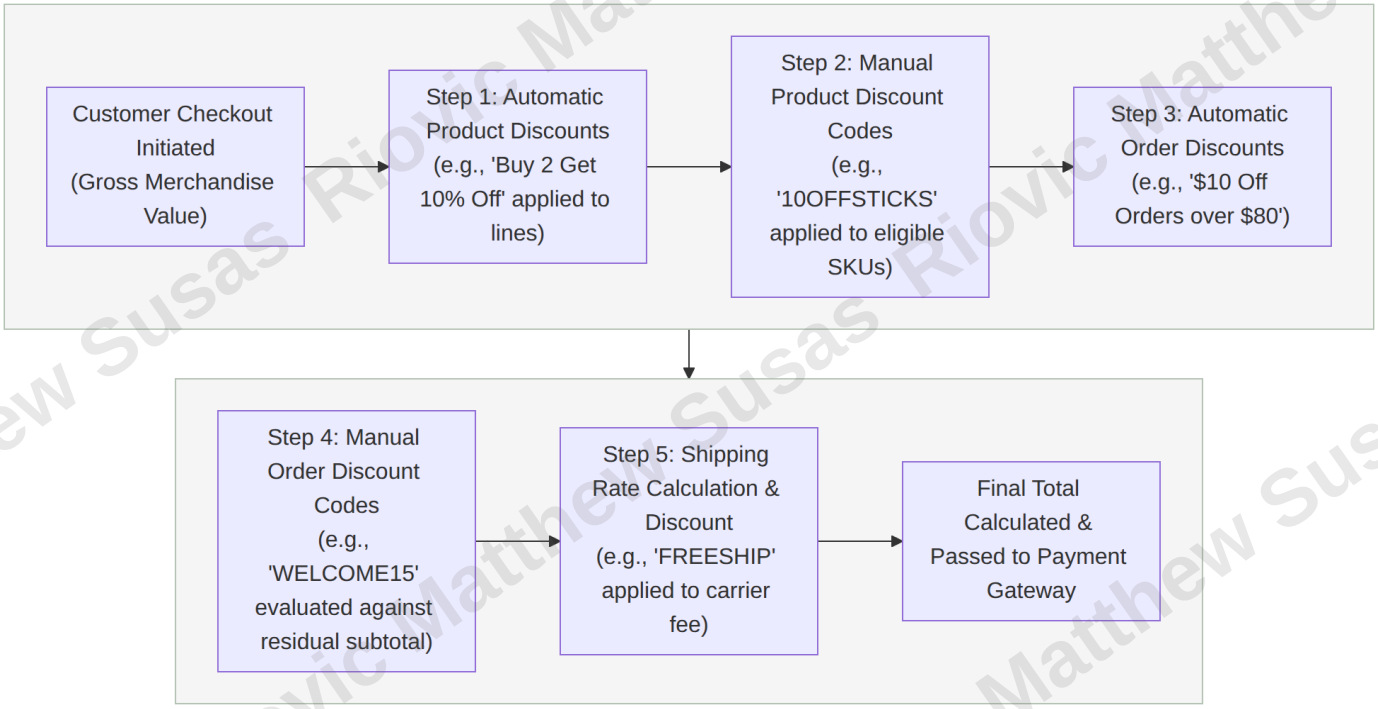
Product & Variant Catalog	Retail Price	Product COGS	3PL Handling (Base + Addl)	Corrugate Packaging	Transcon Carrier Postage	Shopify Gateway Fee (2.9% + $0.30)	Full Cost Baseline	Max Allowable Discount %	Breakeven Floor (CM = 35%)
Focus Stick Packs (30-Pack) `SYN-FSTK-MY` / `BO` / `WB` / `GA`	$44.00	$8.50	$2.75	$0.75	$5.65	$1.58	**$19.23**	**28.5%**	**$31.46**
Adaptogen Tincture (60ml) `SYN-TNCT-AM` / `PM` *(NV Only)*	$38.00	$12.50	$2.75	$0.75	$5.65	$1.40	**$23.05**	**18.0%**	**$31.16**
Precision Insulated Tumbler `SYN-TMBL-MO` / `BW` / `SG`	$32.00	$6.50	$2.75	$1.25	$10.80	$1.23	**$22.53**	**15.0%**	**$27.20**

Product & Variant Catalog	Retail Price	Product COGS	3PL Handling (Base + Addl)	Corrugate Packaging	Transcon Carrier Postage	Shopify Gateway Fee (2.9% + $0.30)	Full Cost Baseline	Max Allowable Discount %	Breakeven Floor (CM = 35%)
Micro-Foam Electric Whisk `SYN-ACC-WHSK` *(NV Only)*	$18.00	$3.20	$2.75	$0.75	$5.65	$0.82	**$13.17**	**8.0%**	**$16.56**
Daily Protocol Starter Bundle `SYN-BNDL-MAO / BAB`	$98.00	$26.50	$3.75	$1.25	$10.80	$3.14	**$45.44**	**32.0%**	**$66.64**

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Key Strategic Takeaways from Margin Modeling:
1. Tumblers (SYN-TMBL-MO) and Whisks (SYN-ACC-WHSK) CANNOT tolerate stand-alone 20% discounts when paired with free shipping, because carrier shipping ($10.80) and 3PL handling absorb up to 50% of retail price.
2. The Daily Protocol Starter Bundle (SYN-BNDL-MAO) has the highest dollar margin buffer ($52.56 pre-discount), safely accommodating the WELCOME15 + FREESHIP stack while maintaining a healthy 35.8% Contribution Margin.
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4. Precedence Hierarchy & Cart Evaluation Order

When a customer arrives at checkout with pre-existing promotional items, automated sales, and enters one or more discount codes, Shopify's calculation engine evaluates the checkout in an immutable, deterministic order:



4.1 Evaluation Rules & Conflict Resolution Principles

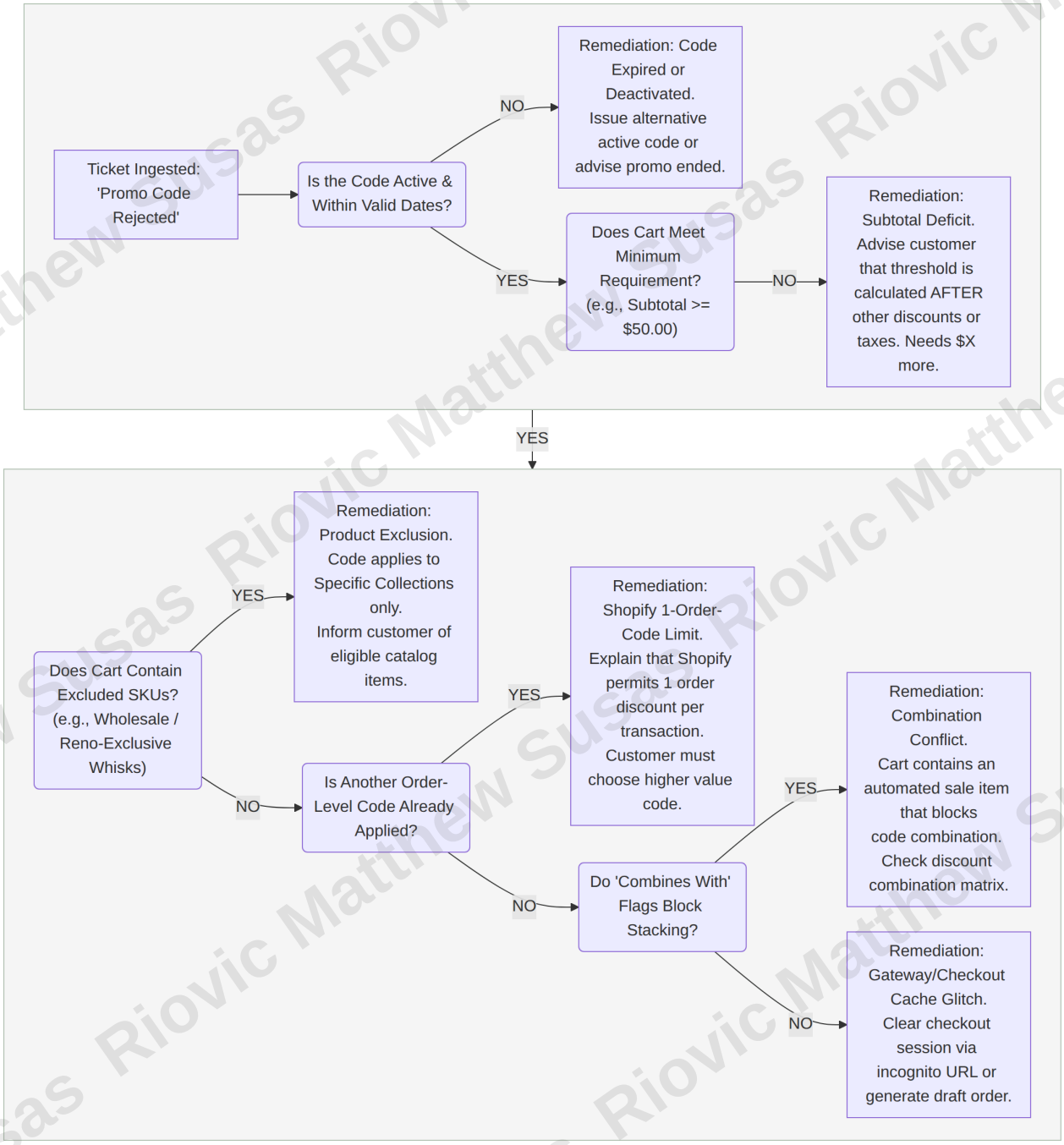
1. **Subtotal Compounding:** Order-level discounts are calculated against the *net line-item subtotal* remaining after product discounts have already been deducted. They never calculate against original gross retail prices.

2. **Code Precedence on Re-Entry:** If a customer enters Order Discount Code A (`WELCOME15`), and subsequently enters Order Discount Code B (`INFLUENCER10`), Shopify **replaces Code A with Code B**. The engine does not aggregate order codes.
3. **Automatic vs. Manual Priority:** If an Automatic Order Discount exists (e.g., "*Memorial Day $15 Off*"), and a customer enters a Manual Order Code (`WELCOME15`), Shopify applies the discount that provides the **greatest financial benefit to the customer**, unless the manual code combination settings explicitly permit stacking with automated rules.

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5. Customer "Discount Code Rejected" Diagnostic Decision Tree

When a customer escalates a complaint stating their promo code was rejected at checkout, TechOps and Tier 2 Support specialists must execute this deterministic diagnostic tree:



5.1 Quick Reference Triage Scripts for Support Specialists

Error Message at Checkout	Diagnostic Root Cause	Customer-Facing Explanation Script
"*This discount has reached its usage limit*"	Usage limit exceeded (Global limit or 1-per-customer limit breached).	"*This exclusive promotional code has reached its maximum authorized redemption limit. We have generated a personal 10% courtesy code for your order today.*"
"*Enter a valid discount code*"	Typo, casing error, or code not published to active sales channel.	"*The code entered was not recognized. Please verify the spelling or note that this code is exclusive to our email subscriber channel.*"
"*This code isn't valid for the items in your cart*"	The discount applies to specific collections (e.g., *Stick Packs*), but the cart contains excluded hardware (*Tumbler* or *Whisk*).	"*This promotion applies exclusively to our botanical dietary formulations (Focus Stick Packs & Tinctures) and cannot be applied to hardware or merchandise.*"
"*Discount cannot be combined with existing promotions*"	Combination flag unchecked; cart already contains an automated sitewide discount.	"*Our checkout automatically applies our best promotional rate. This special code cannot be combined with active sitewide sale pricing.*"

6. Master Interconnectivity & System Architecture Lineage

The mathematical rules and combination governance matrices in this engine connect directly into the operational architecture across all five TechOps domains:

Upstream / Downstream Deliverable	Governing Interface / Node	Interconnectivity Mechanism
SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)	Post-Purchase Line Item Swaps	When customer service swaps a flavor on an order, this engine governs whether original promotional discount codes carry over to the new variant.
Matrix - OrderStatePointOfNoReturn - Master (Domain 1.2)	Financial Ledger & Tax Thresholds	Governs how state sales tax calculations recalculate when discounts adjust net taxable line-item balances.
Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)	Multi-Location Routing Rules	Free shipping threshold discounts (`FREESHIP`) trigger specific fulfillment node allocation across Allentown (`LOC-PA-01`) and Reno (`LOC-NV-02`).
Matrix - OrderRoutingCostOptimization - DecisionEngine (Domain 2.2)	Carrier Zone Rate Benchmark	Supplies the carrier rate baseline ($5.65 to $10.80) used in Section 3.2 unit economics and contribution margin modeling.
Matrix - CatalogDataSchemaAndValidation - Governance (Domain 3.2)	Catalog Pricing & Compare-At Fields	Ensures that `Variant Price` and `Variant Compare At Price` CSV imports do not distort automated discount baseline calculations.
SOP - BusinessLogicAutomationAndWebhooks - Parent (Domain 4.1)	Shopify Flow Automations	Governs Flow Blueprints that tag orders based on discount code application (e.g., tagging influencer referral orders).
SOP - TechnicalDiagnosisAndAuditLogging - Parent (Domain 5.1)	Diagnostic Standards & Cart APIs	Operates as the diagnostic parent specification, providing the AVS/CVV and Order Timeline rules that validate checkout payment completion.
CaseStudy - DiscountCollisionAndCartDiagnostics - Analysis (Domain 5.3)	Empirical Simulation & Checkout Proofs	Provides the empirical testing ground executing live storefront checkouts of `WELCOME15` vs `FREESHIP` and documenting cart network traces.